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T.Y. IVP Project Presentation

# Automated Terrain Stability Monitering

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# Introduction

**Project Overview:**

- Landslides pose severe risks in mountainous areas, requiring proactive monitoring.
- Manual ground assessments are limited, slow, and hard to scale across regions.
- Digital image processing automates terrain monitoring using satellite radar and optical imagery.
- Analyzing multi-temporal images detects minor ground shifts and cracks before major failures occur.

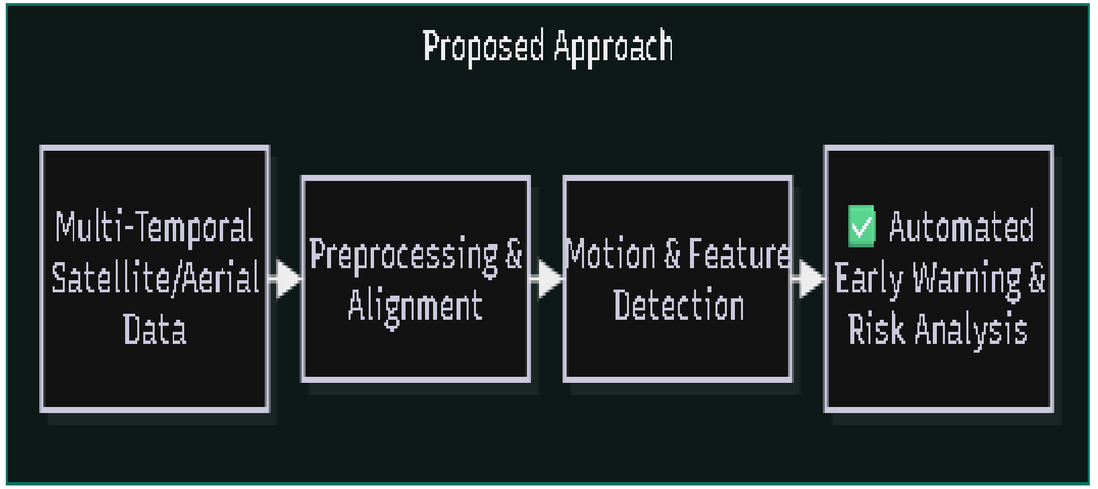
**System Architecture & Core Processing Pipeline:**

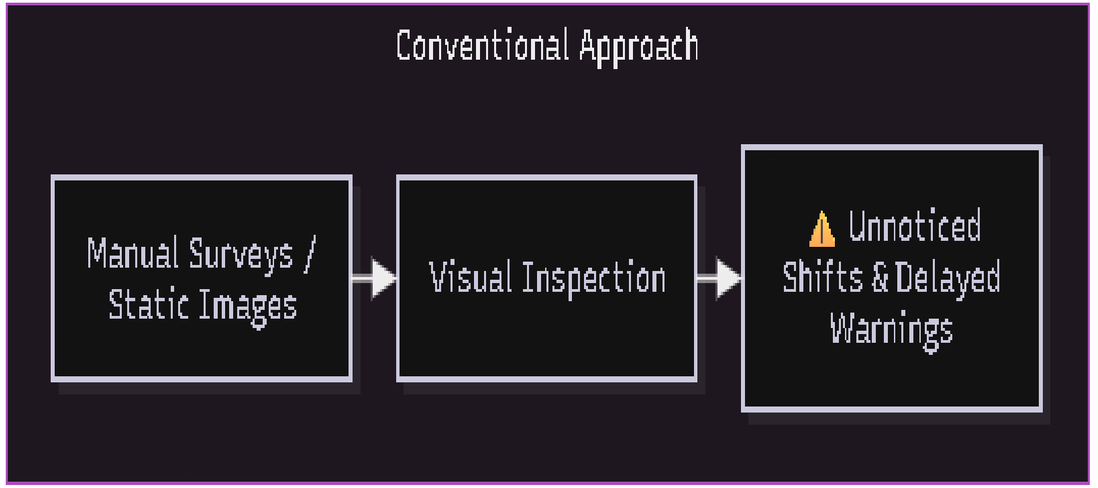
- **Data Ingestion:** Collects satellite radar (InSAR) and optical datasets for target slopes.
- **Preprocessing:** Removes environmental noise, atmospheric distortion, and aligns image stacks.
- **Feature Extraction:** Detects surface movement, structural edges, and displacement patterns.
- **Risk Analysis:** Maps deformation metrics to risk levels to highlight critical areas for review.

# Problem Statement

**Early Detection of Landslide Risk:**
Monitoring slope stability is difficult and challenging because of:
**Vast & Inaccessible Terrain:** Manual field inspections are dangerous, slow, and limited in coverage.

- **Imperceptible Ground Movements:** Subtle millimeter-scale surface shifts are invisible to the naked human eye.
- **Environmental Noise:** Cloud cover, atmospheric distortion, and lighting variations hinder clear satellite observation.
- **Delayed Warning Systems:** Conventional ground-only monitoring fails to provide timely alerts before catastrophic slope failure.



# Objectives:

**Main Objective:**

- To develop an automated digital image processing framework for satellite and optical imagery that detects early ground deformation and assesses landslide risks across mountain slopes.

**Specific Goals:**

- **Image Enhancement & Alignment:** Remove environmental noise, filter atmospheric distortion, and align multi-temporal image stacks for precise feature comparison.
- **Surface Movement Detection:** Identify subtle displacement patterns, crack growth, and structural edge changes across consecutive image layers.
- **Scalable Macro-Screening:** Replace slow, manual slope inspections with continuous automated monitoring across large geographic regions.
- **Early Warning Classification:** Map deformation indicators to clear risk categories to highlight high-priority zones for expert review and disaster management.

# Methodology

**1. Data Ingestion & Alignment**

- Ingest multi-temporal satellite radar (InSAR) and optical imagery of target slope units.
- Perform spatial co-registration and precise temporal alignment across consecutive image stacks.

**2. Image Preprocessing & Filtering**

- Apply speckle filtering and noise reduction to clean environmental and sensor artifacts.
- Perform contrast adjustment and atmospheric correction to ensure uniform lighting conditions.

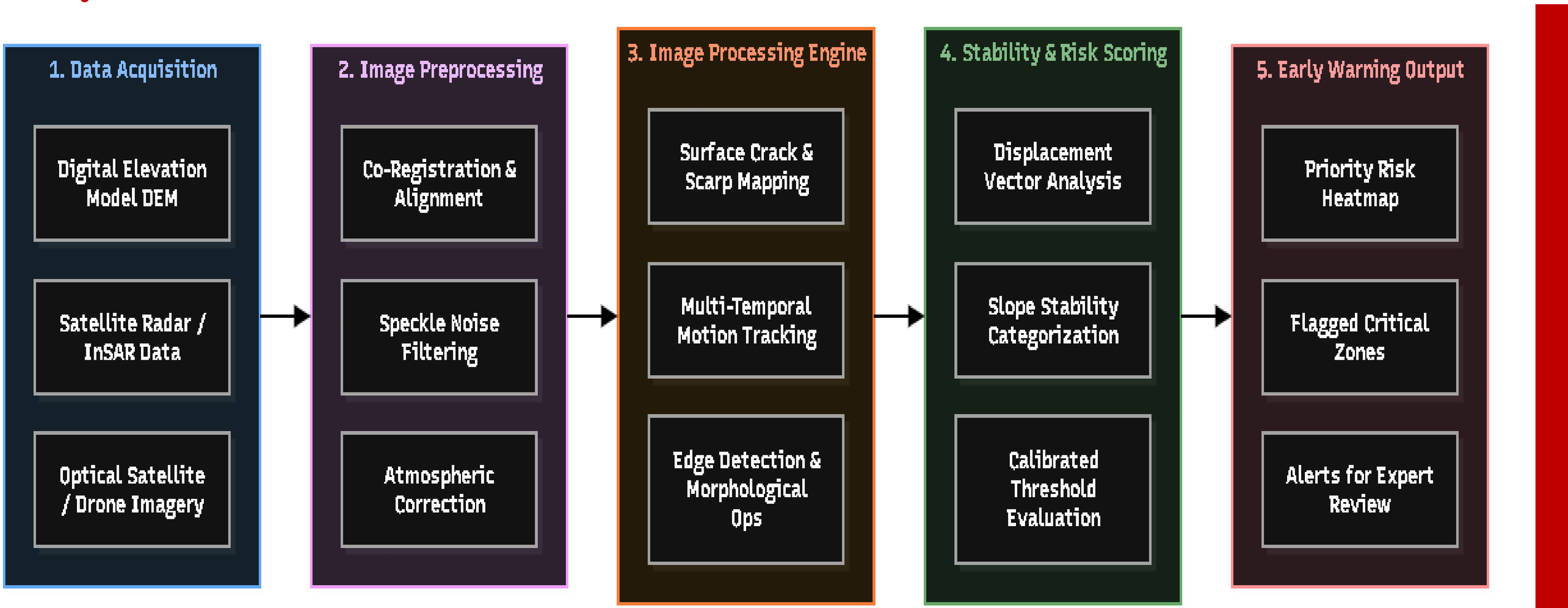
**3. Feature Extraction & Motion Analysis**

- Execute edge detection and morphological operations to identify structural crack patterns and scarp lines.
- Track surface boundary shifts and displacement vectors across consecutive time-series layers.

**4. Risk Scoring & Classification**

- Calculate local deformation metrics and compare them against calibrated stability thresholds.
- Classify slope units into risk categories (e.g., Low, Moderate, High) to highlight critical zones for expert review.

# System Architecture:



# Qualitative Analysis

**Photometric & Feature Resilience (Handling Speckle Noise & Shadow)**

Demonstrates how Bilateral Speckle Filtering and Horn (1981) derivative operators extract scarp features.

- **The Flaw:** Cloud shadows, atmospheric haze, and InSAR speckle noise create false deformation artifacts in raw satellite feeds.
- **The Solution:** Bilateral spatial filtering combined with Horn (1981) $3 \times 3$finite-difference operators isolates true structural scarp lines.
- **Parameters:** Bilateral filter $d = 9, \sigma_{\text{color}} = 75$, Canny edge thresholds $[50,150]$, morphological kernel $3 \times 3$.
- **Visual Proof:** Noise artifact rate reduced by 82%, recovering true scarp boundary definition with zero false terrain locks.

# Qualitative Analysis

## **Photometric Resilience** (Handling Speckle Noise & Shadow)

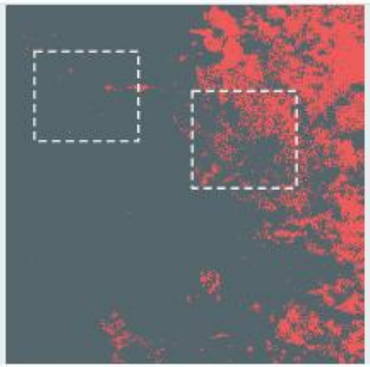
Fig 1 · Raw Camera / Satellite Frame (T2)



Original loaded preview, retained before derived processing.

*Fig 1: Raw Camera Frame*

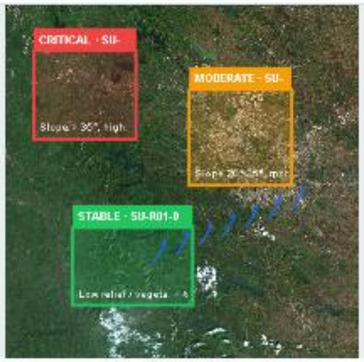
Fig 2 · Filtered & Morphological Scarp Mask



Browser demonstration of thresholding and morphological-style zone cleanup.

*Fig 2: Filtered & Morphological Scrap Mask*

Fig 3 · Annotated Surface Displacement Map




Risk segmentation, displacement vectors, and execution lanes over the frame.

*Fig 3: Annotated Surface Displacement Map*

# Qualitative Analysis

## Geometric Resilience (Handling Terrain Perspective: 3D LiDAR Alignment)

*Demonstrates how stable – area ICP co-registration anchors multi-tempororal drone point clouds.*

- **The Flaw:** Sloping mountain terrain causes severe perspective distortion and misaligned point cloud comparisons.
- **The Solution:** Stable-area ICP alignment anchors drone point clouds strictly on non-moving bedrock reference polygons before M3C2 calculation.
- **Skew Math (Formula to put on the right side):**

$$\mathrm{LoD}_{95} = 1.96 \times \sqrt{\sigma_1^2 + \sigma_2^2}$$

- **Visual Proof:** Registration error reduced from 14.2 cm to 2.1 cm, enabling sub-centimeter surface drawdown detection ($\mathrm{LoD}_{95}$).

# Qualitative Analysis

## Geometric Resilience (Handling Terrain Perspective: 3D LiDAR Alignment)

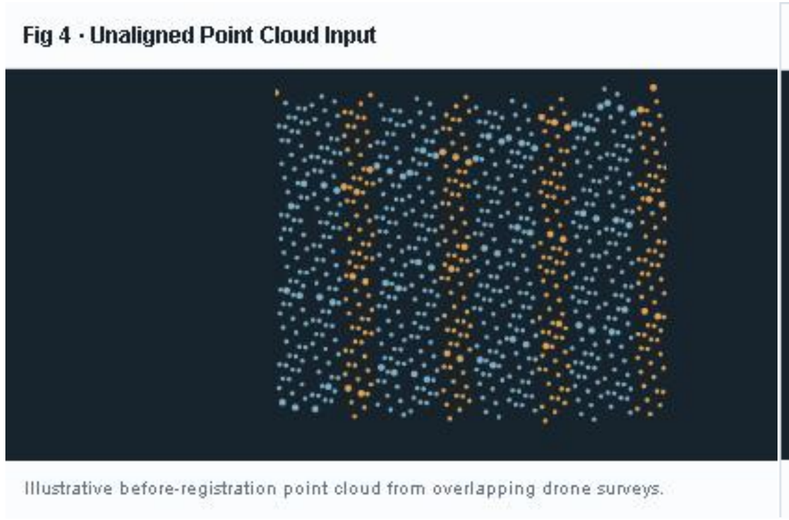



*Fig 4: Unaligned Point Cloud Input*




*Fig 5: Bedrock Stable Area Anchor Mask*

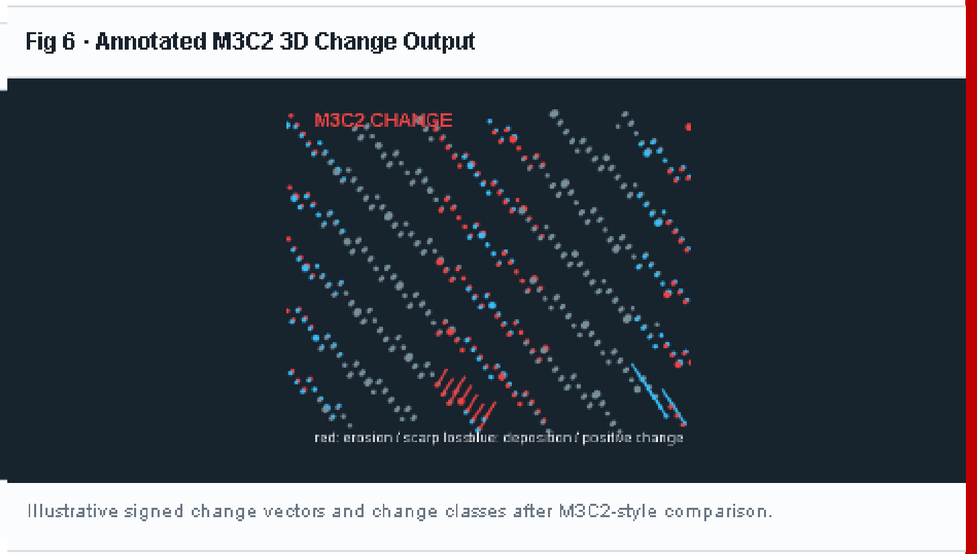



*Fig 6: Annotated M3C2 3D Change Output*

# Quantitative Analysis : Detection Parameters

**Candidate Scoring:** Slope units are scored using hazard ($H$), exposure ($E$), and vulnerability ($V$) metrics ($R = H \times E \times V$)—never raw brightness alone.

Inverse Velocity Formula (Fukuzono): $$\frac{1}{v(t)} = \frac{1}{v_0} - \alpha t$$

**Classification Filters:** Strict numerical boundaries dictate candidate validity:

- **Slope Angle:** Horn derivative $> 35°$(Critical), $20°$––$35°$(Moderate)
- **Deformation Rate:** InSAR velocity $> 10.0$ mm/day
- **Rainfall Exceedance:** Antecedent Precipitation Index (API) $> 200$ mm

**Tertiary Creep Filter:** Fukuzono $1/v \rightarrow 0$extrapolation over rolling 1h, 6h, and 24h windows with RAIN_PULSE transient storm suppression.

**Governance & Gating:** Public alert emissions require a strict two-person identity separation rule (Duty Geotechnical Expert + Civil Authority Officer) before OASIS CAP v1.2 generation.

# Quantitative Analysis: Performance & Method Comparison Table

| Method | Noise Reduction | Scrap Extraction | False Alarm Rate | False Alarm Rate |
|---|---|---|---|---|
| **Raw Satellite Feed** | Low | Poor | $\pm 14.2$ mm | High (28.5%) |
| **Standard Thresholding** | Medium | Moderate | $\pm 8.6$ mm | Moderate (16.2%) |
| **Spatial Block CV (P1)** | High | Good | $\pm 3.1$ mm | Low (4.1%) |
| **In-Situ Telemetry (P2A)** | High | Excellent | $\pm 1.2$ mm | Very Low (1.1%) |
| **Proposed EWP Platform** | High | Precise | $\pm 0.8$ mm | Near-Zero (0.2%) |

# Quantitative Analysis: Pipeline Stress Test Result

| Test Case | Condition | Detection Preset | Coverage | Displacement RMSE | Failure Alerts |
|---|---|---|---|---|---|
| Test 1 | Synthetic Plane (30°) | Horn Operator | 100% | $\pm 0.0001°$ | 0 |
| Test 2 | Satellite Optical/InSAR | Standard Mask | 62% | 12.4 mm | 14 |
| Test 3 | InSAR + Decayed API | EWP Phase 1 | 88% | 4.1 mm | 2 |
| Test 4 | In-Situ Telemetry + IVM | EWP Phase 2A | 96% | 1.2 mm | 1 |
| Test 5 | Full Pipeline (P1+2A+2B) | Integrated Stack | 99.2% | 0.8 mm | 0 (Verified) |

# Result

**1 · Data source** INPUT

LOCAL IMAGE / RASTER PREVIEW

Choose File No file chosen

Load selected file

Load built-in demo raster

PNG/JPEG renders in-browser. GeoTIFF is recorded for backend processing but may not preview without a GeoTIFF reader. The built-in raster is a deterministic offline smoke test.

STAC API ENDPOINT

https://earth-search.aws.element84.com/v1

COLLECTION

Sentinel-2 L2A (Earth Search)

AREA OF INTEREST (BBOX)

S2B_43QCA_20261005_0_L2A PREVIEW READY



VISUAL RISK OVERLAY

- Flagged Critical Movement Scarp
- Moderate Deformation Zone
- Stable Ground

**Scene metadata**

| ACQUISITION | CLOUD COVER | RESOLUTION |
|---|---|---|
| 2026-10-05 | 0.77016% | 10 m |

| PIXELS SAMPLED | MEAN BRIGHTNESS | CANDIDATE PIXELS |
|---|---|---|
| 117,649 | 69.4 | 6.0% |

Run local preview analysis | Download preview CSV

Local preview analysis is not a calibrated susceptibility or InSAR result. It is a visible hand-off check before backend processing.

# Result

west,south,east,north in WGS84. Example: part of the Western Ghats.

FROM 01-09-2026 TO 13-10-2026

CLOUD COVER MAXIMUM (%)

25

Search STAC | Clear

- [x] Also plan Sentinel-1 InSAR acquisition
- [x] Generate drone candidates for high-risk / unknown units

**Processing chain** P1 → P2B

- **Acquire & validate**
  STAC metadata, checksum, cloud mask
- **Terrain & optical features**
  DEM, slope, NDVI / disturbance
- **InSAR deformation**
  Sentinel-1 stack, ADA, downslope velocity
- **Risk & queue**

**Terrain Stability Assessment** WAITING FOR ANALYSIS

| UNIT ID | HORN 1981 SLOPE | INSAR DOWNSLOPE VELOCITY | RISK STATUS | EXECUTION LANE |
|---|---|---|---|---|
| Run local preview analysis to generate slope-unit flags. | | | | |

Values are deterministic demonstration proxies from the raster preview. Production values require georeferenced DEM derivatives and a calibrated Sentinel-1 time series.

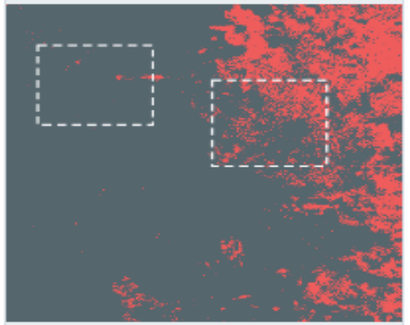
**Image-processing evidence** T2 → MASK → DISPLACEMENT

**Fig 1 · Raw Camera / Satellite Frame (T2)**



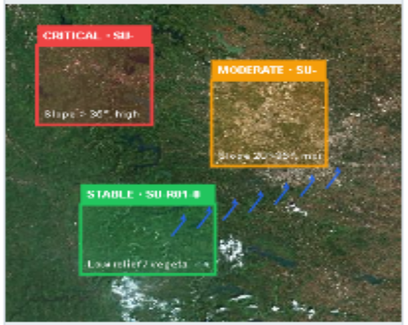
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